PROJECT CASE STUDY

OTA Support

Technician Resource Portal

I built otasupport.com to help technicians find service manuals, technical documents, and other reference materials independently, while reducing routine access requests to the warranty department.

Why the project was needed

The resource library already existed in Google Drive, but technicians relied on the warranty department to arrange access. Staff handled permissions manually alongside their warranty responsibilities, and technicians depended on that intervention before they could use the materials.

Finding information was another difficulty. A shared folder provided access to the files, but navigating a large library through Google Drive was cumbersome. The project needed to improve both access to published resources and document discovery.

What I built

The portal presents the library through search and folder navigation. Technicians can search document names and folder paths or browse the existing hierarchy. Selecting a resource opens it directly in Google Drive, which continues to handle file access and viewing.

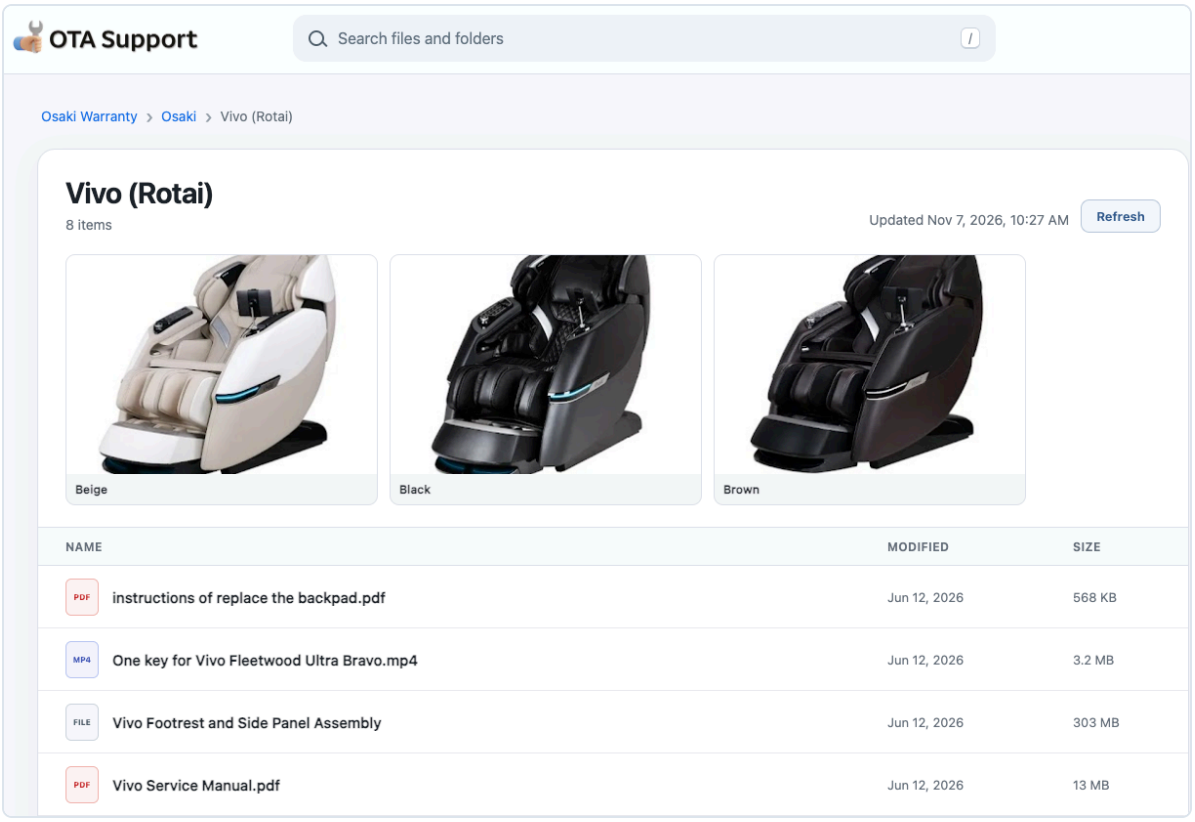


Figure 1. Technicians can search the library, browse its folders, and open a manual or other service resource.

Publishing and architecture

Keeping the library current

I designed the publishing workflow around the tools staff already used. Google Drive remains the source library: staff organize and update documents in familiar folders, and the portal builds its catalog from that structure. This keeps the website aligned with the library without requiring a second collection of documents or a separate publishing system.

Every folder named **In Production (Hold)** and its contents are excluded from the portal's catalog. This gives staff a designated place for unfinished material that should not appear in search or navigation. Google Drive sharing settings continue to govern access to the files.

How the application works

The application is written in TypeScript. A Node.js web application runs on an EC2 instance, S3 stores the prepared catalog, and a Lambda function reads Google Drive to generate it. EventBridge Scheduler runs the scan daily; the portal's Refresh action can invoke the same process when a change needs to appear sooner.

The catalog contains document names, folder relationships, and links to the original files. Manuals, videos, and other resources remain in Google Drive, which provides document viewing and media delivery.

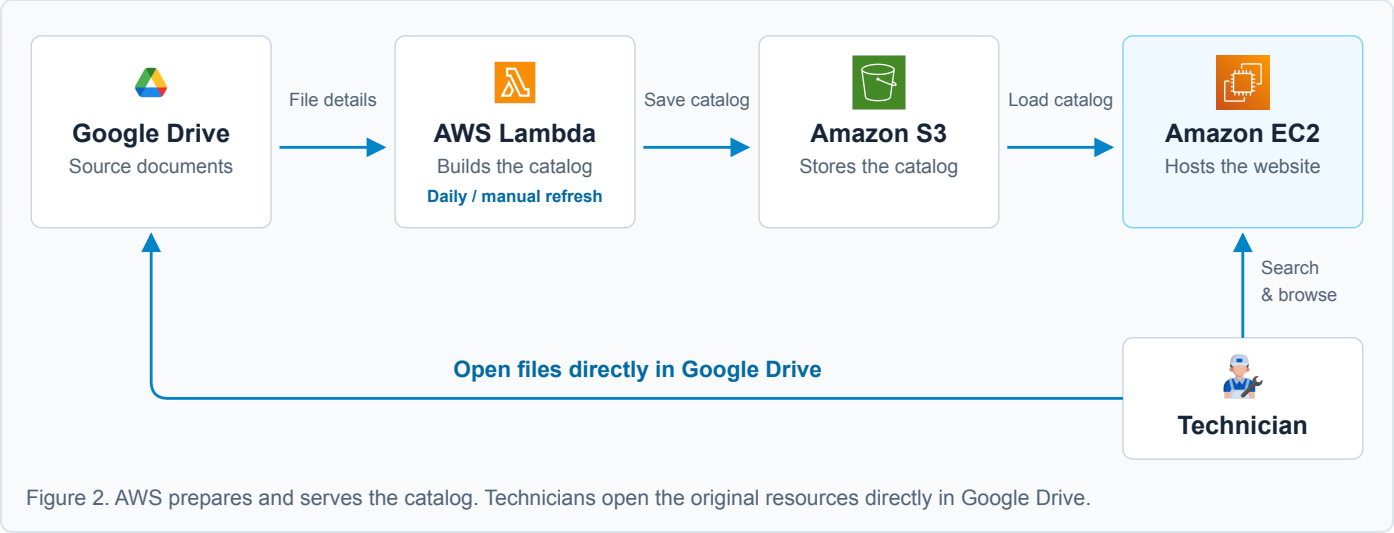


Figure 2. AWS prepares and serves the catalog. Technicians open the original resources directly in Google Drive.

Result

The earlier version recorded **4,573 visitors** and **13,200 page views** over a 12-month period. The portal gives technicians a central place to locate published resources independently, while staff retain a familiar process for keeping the library current.